Cyber Threat Monitoring Level I (CTM-L1)

Institutional Assessment

Specific Instructions (Demonstration)

"Network Intrusion Detection and Initial Incident Response Assessment"

Objectives

At the end of this assessment, trainees must be able to:

1. **Analyze captured given captured data**
2. **Identify cyber threats** from packet-level evidence
3. **Interpret suspicious activities** and correlate indicators
4. **Document findings** clearly and accurately
5. **Develop an incident ticket report** for escalation to Level II (Incident Handling)
6. **Provide evidence-based conclusions**, including screenshots and analysis

Assessment Scenario

Scenario:

You are a **Level I Cyber Threat Monitoring Analyst** assigned to the Security Operations Center (SOC).

During routine monitoring, your SIEM alerts show unusual traffic from multiple internal hosts. A packet capture (PCAP) file was generated and provided to you for analysis.

Initial alerts indicate:

- Multiple failed logins
- Suspicious outbound connections
- Indicators of ransomware behavior
- Possible Command & Control (C2) communications
- Potential DDoS activity and port scanning

Your task is to analyze the packet capture using Wireshark, identify threats, and prepare an escalation report for Level II Incident Handlers.

Provided Sample Packet Data

No	Date	Day	Time	Source	Destination	Protocol	Length	Info
1	2026-05-29	Friday	09:30:00	192.168.1.10	192.168.1.1	TCP	74	SYN Packet (Port 22)
2	2026-05-29	Friday	09:30:00	192.168.1.10	192.168.1.1	TCP	74	SYN Packet (Port 23)
3	2026-05-29	Friday	09:30:00	192.168.1.10	192.168.1.1	TCP	74	SYN Packet (Port 80)
4	2026-05-29	Friday	09:30:00	192.168.1.20	192.168.1.5	SSH	120	Failed Login attempt
5	2026-05-29	Friday	09:30:00	192.168.1.20	192.168.1.5	SSH	120	Failed Login attempt
6	2026-05-29	Friday	09:30:00	192.168.1.20	192.168.1.5	SSH	120	Failed Login attempt
7	2026-05-29	Friday	09:30:01	192.168.1.30	45.33.32.156	HTTP	512	GET /malware.exe (Suspicious URL)
8	2026-05-29	Friday	09:30:01	192.168.1.30	45.33.32.156	HTTP	1024	File Download (Possible Malware)
9	2026-05-29	Friday	09:30:02	192.168.1.40	8.8.8.8	DNS	90	Query strange-domain.xyz
10	2026-05-29	Friday	09:30:02	192.168.1.40	185.220.101.1	TCP	200	Outbound connection (Tor Node)
11	2026-05-29	Friday	09:30:03	192.168.1.50	10.10.10.10	SMB	300	File encrypted extension .locked
12	2026-05-29	Friday	09:30:03	192.168.1.50	10.10.10.10	SMB	320	Ransomware note delivery
13	2026-05-29	Friday	09:30:04	192.168.1.60	192.168.1.255	UDP	1500	High volume broadcast (DDoS)
14	2026-05-29	Friday	09:30:04	192.168.1.61	192.168.1.255	UDP	1500	Flood traffic (DDoS)
15	2026-05-29	Friday	09:30:05	192.168.1.70	103.21.244.0	TCP	200	C2 Communication beacon
16	2026-05-29	Friday	09:30:05	192.168.1.70	103.21.244.0	TCP	200	Encrypted callback traffic
17	2026-05-29	Friday	09:30:06	192.168.1.80	192.168.1.1	TCP	74	Rapid SYN scanning (Port Scan)
18	2026-05-29	Friday	09:30:06	192.168.1.80	192.168.1.2	TCP	74	Rapid SYN scanning (Port Scan)
19	2026-05-29	Friday	09:30:06	192.168.1.90	52.12.34.56	HTTP	800	POST suspicious data upload
20	2026-05-29	Friday	09:30:07	192.168.1.91	192.168.1.10	ICMP	100	Ping sweep
21	2026-05-29	Friday	09:30:07	192.168.1.92	192.168.1.255	UDP	1500	Broadcast amplification
22	2026-05-29	Friday	09:30:08	192.168.1.93	45.33.32.156	HTTP	900	Repeated malware callback
23	2026-05-29	Friday	09:30:08	192.168.1.94	8.8.4.4	DNS	95	Query malicious-site.biz
24	2026-05-29	Friday	09:30:09	192.168.1.95	10.10.10.10	SMB	310	Suspicious file rename
25	2026-05-29	Friday	09:30:09	192.168.1.96	103.21.244.0	TCP	210	C2 data exchange
26	2026-05-29	Friday	09:30:10	192.168.1.97	192.168.1.1	TCP	74	Port scan SYN
27	2026-05-29	Friday	09:30:10	192.168.1.98	192.168.1.5	SSH	120	Repeated login attempt
28	2026-05-29	Friday	09:30:11	192.168.1.99	192.168.1.255	UDP	1500	DDoS burst traffic

Threat Indicators Included

Trainees should identify:

- Failed Logins → possible **Brute Force Attack**
- Suspicious URLs → malware download
- Outbound Traffic → data exfiltration / suspicious IPs
- Ransomware Indicators → encrypted files & ransom notes
- C2 Communications → beaconing behavior
- Port Scanning → reconnaissance
- DDoS activity → flood traffic

REQUIRED OUTPUTS

1. Investigation Documentation (PDF)

Must include:

- Screenshots (Wireshark filters, packets, flows)
- Findings (identified threats)
- Analysis (interpretation of packet data)
- Ticket Evidence
- Escalation Evidence
- Final Conclusion

2. Incident Ticket Evidence

Provide screenshots showing:

- Created ticket ID
- Assigned analyst (Use your handle name)
- Severity level (Low / Medium / High / Critical)
- Notes entered

3. Escalation Evidence

Include:

- Screenshot of escalation action
- Reason for escalation:
    - Example:
        - “Multiple indicators of compromise detected including ransomware activity and C2 communications”
- Severity justification:
    - Example:
        - “Critical due to active data encryption and external communication”

4. Final Incident Report

Must answer:

What happened?

- Summary of attack (multi-stage attack: scanning → brute force → malware → ransomware)

How was the attack detected?

- Alerts and packet evidence from Wireshark

What evidence supports the attack?

- Packet logs (failed logins, suspicious downloads, encrypted SMB traffic, beaconing)

What actions should be recommended?

- Isolate infected hosts
- Block malicious IPs/domains
- Reset compromised credentials
- Initiate malware removal
- Monitor C2 communications
- Apply patches and security controls

Instructor Notes

- This is a **simulation-based SOC exercise**
- Focus is on **analysis, documentation, and escalation**

- No remediation required at Level I (only reporting)